

## Multiple Choice

1. **Answer:** B

*Options:* A) PROM; B) MROM; C) DRAM; D) EPROM; E) SRAM

*Note:* Correct option: B - MROM

2. **Answer:** A

*Options:* A) 0.3 msec; B) 0.8 msec; C) 0.6 msec; D) 0.25 msec; E) 0.1 msec

*Note:* Correct option: A - 0.3 msec

3. **Answer:** A

*Options:* A)  $-e^{-t}-e^2 t + 2 t e^2 t - (t^2 / 2)e^2 t$ ; B)  $e^{-t} - 2e^2 t + 3 t e^2 t - (t^3 / 6)e^2 t$ ; C)  $-e^{-t} + 3e^2 t - t^2 e^2 t$ ; D)  $-e^{-t} + 2e^2 t - 2 t e^2 t + (t^3 / 3)e^2 t$ ; E)  $e^{-t} + e^2 t - 2 t e^2 t + (t^2 / 2)e^2 t$

*Note:* Correct option: A -  $-e^{-t}-e^2 t + 2 t e^2 t - (t^2 / 2)e^2 t$

4. **Answer:** C

*Options:* A)  $C = (\epsilon^2 S / d)$ ; B)  $C = (/d^2)$ ; C)  $C = (/d)$ ; D)  $C = (\epsilon / dS)$

*Note:* Correct option: C -  $C = (/d)$

5. **Answer:** E

*Options:* A) 150; B) 100; C) 200; D) 5; E) 10

*Note:* Correct option: E - 10

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '15.42 psia'.

7. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* Actual answer is ' $a_n = [(2 ca) / \sqrt{a^2 + b^2}]$ ,  $a_t = [\{2 c(b - ct)\} / \sqrt{a^2 + b^2}]$ '.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: ' $ds = C_p (dT / T) - (R / J)(dp / p)$ '.

10. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: ' $k / (s^2 + k^2)$ , for  $s > 0$ '.

## Long Form Response

11. **Answer:** 0.3. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 6. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*